A candidate proof of the Gatzweiler–Krattenthaler positivity conjecture

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Abstract

Gatzweiler and Krattenthaler conjectured that a polynomial quotient of two Gaussian binomial coefficients with the same upper index has nonnegative coefficients. We present a candidate proof for the complete parameter range. After cancellation, polynomiality gives divisibility-count inequalities for two separated integer intervals of equal length. We prove that every subset of the lower interval generates at least as many elements of the upper interval in its additive semigroup. The criterion of Gatzweiler, Levicán-Santibáñez and Yoshida then supplies a regular sequence whose Hilbert series is the required quotient. The central lemma uses a mixed sumset modulo the largest generator not exceeding the interval length. Kneser's theorem, together with separate capacities for the lower and upper generators, handles all stabilizers and all collisions of residues. Two-generator representations treat subsets near the upper endpoint. A second sumset estimate handles small lower indices, and a necessary inequality from polynomiality leaves only two explicit boundary families. The proof has no finite parameter cutoff; exact computations are supplied as supplementary checks.

Preprint; not externally peer reviewed. AI use and verification are described in the appendix.

Introduction and proof strategy

The Gaussian binomial coefficient is a polynomial with nonnegative integer coefficients, but quotients of such polynomials need not preserve coefficientwise nonnegativity. Gatzweiler and Krattenthaler [1] proposed that polynomiality is sufficient when the two Gaussian binomial coefficients have the same upper index. Their paper treats the cases $l = 0, 1, 2$ and $k-l = 0, 1, 2, 3$, and reports exact verification for $n \leq 400$. Gatzweiler, Levicán-Santibáñez and Yoshida [2] subsequently established a sufficient condition using regular sequences in weighted polynomial rings. In the normalized range $1 \leq l < k \leq n/2$, their Corollary 1.7 proves the conjecture when

$$2k \left\lfloor \frac{k-1}{2} \right\rfloor + k - l < n,$$

and hence when $n \geq k^2$.

The present argument uses the semigroup criterion of [2] directly. It establishes that criterion throughout the normalized parameter range. Besides the regular-sequence criterion and its Hilbert-series consequence, the only external theorem used is Kneser's addition theorem, in the form recorded by DeVos [3]. All representation estimates and all parameter reductions needed below are proved in the manuscript.

Proof strategy. Write $d = k - l$, and cancel the factorials in the Gaussian binomial quotient. The numerator exponents form an interval A of length d lying above the denominator interval

$B = [l+1, k]_{\mathbb{Z}}$. Polynomiality says that A contains at least as many multiples of every positive integer as B does. The task is to show

$$|A \cap \langle I \rangle| \geq |I| \qquad (\varnothing \neq I \subseteq B),$$

where $\langle I \rangle$ is the additive semigroup generated by I.

If I has a generator at most d, a mixed sumset yields the desired inequality. Its two summands record all generators and the generators at most d, respectively. Keeping these two capacities separate is what allows the periodic and noninjective residue cases to be counted correctly. If all generators exceed d, collisions of direct multiples can instead be converted into many two-generator representations when $k \geq 3l-1$. A further sumset argument covers $l \leq 3(\lfloor n/k \rfloor - 1)$. Polynomiality forces the remaining parameters onto two boundary families, where direct injections finish the proof.

Reading guide. Section 1 states the results and the interval reduction. Section 2 records the external inputs and an elementary two-generator bound. Sections 3 and 4 prove the mixed-sumset and upper-endpoint lemmas. Sections 5 and 6 establish the small-index criterion. Sections 7 and 8 identify and settle the boundary families, and Section 9 completes both the semigroup statement and the original conjecture.

1 Main results and the interval reduction

For nonnegative integers n and $0 \leq j \leq n$, put

$$[n]_q! = \prod_{i=1}^{n} \frac{1-q^i}{1-q}, \qquad [0]_q! = 1, \qquad \begin{bmatrix} n \\ j \end{bmatrix}_q = \frac{[n]_q!}{[j]_q!\,[n-j]_q!}. \tag{1.1}$$

All polynomials and rational functions below have the indeterminate q. The notation $\mathbb{Z}_{\geq 0}[q]$ denotes polynomials with nonnegative integer coefficients.

Theorem 1.1 (The proposed positivity theorem)**.** *Let n, k, l be integers with $0 \leq k, l \leq n$. If*

$$\frac{\begin{bmatrix} n \\ k \end{bmatrix}_q}{\begin{bmatrix} n \\ l \end{bmatrix}_q} \tag{1.2}$$

is a polynomial in q, then it belongs to $\mathbb{Z}_{\geq 0}[q]$.

We will prove a stronger semigroup assertion. For integers u, v, write $[u, v]_{\mathbb{Z}} = \{x \in \mathbb{Z} : u \leq x \leq v\}$; this set is empty when $u > v$. For a finite nonempty set I of positive integers, define

$$\langle I \rangle = \left\{ \sum_{b \in I} t_b b : t_b \in \mathbb{Z}_{\geq 0} \right\}. \tag{1.3}$$

In particular, $0 \in \langle I \rangle$.

Theorem 1.2 (Interval semigroup condition)**.** *Let $0 \leq l < k \leq n/2$, and put*

$$d = k - l, \qquad N = n - k + 1, \qquad U = n - l, \qquad A = [N, U]_{\mathbb{Z}}, \qquad B = [l+1, k]_{\mathbb{Z}}. \tag{1.4}$$

Suppose that, for every positive integer s,

$$C_A(s) := |A \cap s\mathbb{Z}| \;\geq\; C_B(s) := |B \cap s\mathbb{Z}|. \tag{D}$$

Then, for every nonempty subset $I \subseteq B$,

$$|A \cap \langle I \rangle| \geq |I|. \tag{H}$$

The two intervals in (1.4) have the same cardinality d, and $N > k$. Factorial cancellation gives

$$P_{n,k,l}(q) := \frac{\begin{bmatrix} n \\ k \end{bmatrix}_q}{\begin{bmatrix} n \\ l \end{bmatrix}_q} = \frac{\prod_{a\in A}(1-q^a)}{\prod_{b\in B}(1-q^b)}. \tag{1.5}$$

For $s \geq 2$, the exponent of the cyclotomic polynomial $\Phi_s(q)$ in this rational function is $C_A(s) - C_B(s)$. The factors with $s = 1$ cancel because $|A| = |B|$. Consequently

$$P_{n,k,l}(q) \text{ is a polynomial} \quad \Longleftrightarrow \quad \text{(D)}. \tag{1.6}$$

This also shows that a polynomial quotient automatically lies in $\mathbb{Z}[q]$: it is a product of cyclotomic polynomials with nonnegative integer exponents. Thus the phrase *is a polynomial* in Theorem 1.1 does not require an additional integrality assumption.

For later use, the counts are

$$C_A(s) = \left\lfloor \frac{U}{s} \right\rfloor - \left\lfloor \frac{N-1}{s} \right\rfloor, \qquad C_B(s) = \left\lfloor \frac{k}{s} \right\rfloor - \left\lfloor \frac{l}{s} \right\rfloor. \tag{1.7}$$

In particular, (D) implies that every $b \in B$ has a positive multiple in A.

To reduce Theorem 1.1 to the normalized range, use $\begin{bmatrix} n \\ j \end{bmatrix}_q = \begin{bmatrix} n \\ n-j \end{bmatrix}_q$ to move both lower indices into $[0, \lfloor n/2 \rfloor]_{\mathbb{Z}}$. Equal indices give the quotient 1. The degree $j(n-j)$ is strictly increasing on these integer indices. If the numerator has the smaller lower index, the rational function has negative degree and therefore cannot be a nonzero polynomial. All remaining nontrivial cases have $0 \leq l < k \leq n/2$.

Symbol	Meaning in the normalized range
$d = k - l$	Common number of numerator and denominator exponents
$A = [N, U]_{\mathbb{Z}}$	Numerator exponents, with $N > k$ and $U - N + 1 = d$
$B = [l+1, k]_{\mathbb{Z}}$	Denominator exponents
I	An arbitrary nonempty subset of B
$\langle I \rangle$	Nonnegative integer combinations of the elements of I
$t = \lfloor n/k \rfloor$	The layer parameter, always at least 2

2 Semigroup, sumset, and representation tools

2.1 The regular-sequence criterion

We use the following consequence of Propositions 3.3 and 3.6 of Gatzweiler, Levicán-Santibáñez and Yoshida [2].

Theorem 2.1 (Semigroup criterion)**.** *Let A and B be finite sets of positive integers with $|A| = |B|$. If $|A \cap \langle I \rangle| \geq |I|$ for every nonempty $I \subseteq B$, then*

$$\frac{\prod_{a\in A}(1-q^a)}{\prod_{b\in B}(1-q^b)} \in \mathbb{Z}_{\geq 0}[q].$$

More precisely, in the weighted polynomial ring $R = \mathbb{C}[x_b : b \in B]$, with $\deg x_b = b$, there is a homogeneous regular sequence $(f_a)_{a\in A}$ with $\deg f_a = a$, and the displayed quotient is $\mathrm{Hilb}(R/(f_a : a \in A); q)$.

The coefficient of q^j in this Hilbert series is the dimension of the degree-j component of the quotient ring. Equal numbers of variables and regular-sequence elements make that quotient finite dimensional. This explains both nonnegativity and polynomiality in the conclusion. Theorem 1.2, together with this criterion and (1.6), will imply Theorem 1.1.

2.2 Kneser's theorem and cyclic stabilizers

For subsets S, T of an abelian group, write $S+T = \{s+t : s \in S,\ t \in T\}$, and let $pS = S+\cdots+S$ denote the p-fold sumset. The stabilizer of a set Q is $\mathrm{Stab}(Q) = \{h : Q + h = Q\}$.

Theorem 2.2 (Kneser)**.** *Let S, T be nonempty finite subsets of an abelian group, and let $H = \mathrm{Stab}(S+T)$. Then*

$$|S+T| \geq |S+H| + |T+H| - |H|. \tag{2.1}$$

See [3, Theorem 1]. If $S, T, S+T$ occupy respectively z, w, v cosets of H, then $S+T$ is a union of full H-cosets and

$$v \geq z + w - 1. \tag{2.2}$$

We will also use

$$|pS| \geq p|S+H| - (p-1)|H|, \qquad H = \mathrm{Stab}(pS). \tag{2.3}$$

To justify this form, pass to the quotient by H. The image of pS has trivial stabilizer. The stabilizer of every intermediate sumset jS in the quotient is contained in that of pS, because adding $(p-j)S$ preserves every period of jS. Thus each intermediate sumset is aperiodic. Applying (2.1) successively in the quotient gives $|p\overline{S}| \geq p|\overline{S}| - (p-1)$, which lifts to (2.3).

Every subgroup of $\mathbb{Z}/m\mathbb{Z}$ has the form

$$H = \{0, g, 2g, \ldots, m-g\}, \qquad g \mid m. \tag{2.4}$$

Its order is $h = m/g$, and its cosets are indexed by residues modulo g. A set stabilized by H therefore contains every residue modulo m lying in any of the modulo-g classes it occupies.

2.3 An elementary two-generator bound

Lemma 2.3. *Let a, b be distinct positive integers. Put $g = \gcd(a,b)$, $L = ab/g$, and $F = L-a-b$. Every nonnegative multiple of g strictly greater than F belongs to $\langle a, b\rangle$.*

Proof. Write $a = ug$ and $b = vg$, with $\gcd(u,v) = 1$. For an integer $w > uv - u - v$, choose $j \in \{0, 1, \ldots, u-1\}$ such that $jv \equiv w \pmod{u}$. Then

$$w - jv > uv - u - v - (u-1)v = -u.$$

Since $w - jv$ is a multiple of u, it is nonnegative. Hence $w = (w-jv) + jv \in \langle u, v\rangle$. Scaling by g proves the assertion. □

3 The mixed-sumset lemma

Throughout this section, A and B are as in (1.4), with $N > k$ and (D). The following lemma treats every subset containing a generator no larger than the common interval length.

Lemma 3.1 (Mixed sumsets)**.** *If $\varnothing \neq I \subseteq B$ and $I \cap [1,d]_{\mathbb{Z}} \neq \varnothing$, then $|A \cap \langle I\rangle\,| \geq |I|$.*

Proof. Define

$$I_{\mathrm{low}} = I \cap [1,d]_{\mathbb{Z}}, \qquad m = \max I_{\mathrm{low}}, \qquad M = \max I. \tag{3.1}$$

Then $l < m \leq d$, and

$$I_{\mathrm{low}} \subseteq [l+1, m]_{\mathbb{Z}}, \qquad I \setminus I_{\mathrm{low}} \subseteq [d+1, k]_{\mathbb{Z}}. \tag{3.2}$$

The second interval has length l. There are no elements of I in $[m+1, d]_{\mathbb{Z}}$; this uses the maximal choice of m.

Residues and their lifts. In $\mathbb{Z}/m\mathbb{Z}$, put

$$S = I \bmod m, \qquad T = I_{\text{low}} \bmod m, \qquad Q = S + T. \tag{3.3}$$

Both S and T contain 0, since $m \in I_{\text{low}}$. Suppose $y \in A$ and $y \bmod m \in Q$. Represent its residue as a sum of one residue from S and one from T. For a nonzero residue choose a corresponding generator, and for a zero residue choose the empty sum. The resulting integer z belongs to $\langle I \rangle$ and satisfies

$$0 \le z \le M + m, \qquad z \equiv y \pmod{m}. \tag{3.4}$$

If $z > y$, the congruence would give $z \ge y + m \ge N + m > M + m$, a contradiction. Therefore $y = z + jm$ for some $j \ge 0$, and $y \in \langle I \rangle$. It suffices to count the elements of A whose residues lie in Q.

Capacities of occupied cosets. Let $H = \mathrm{Stab}(Q)$, write it as in (2.4), and put $h = m/g$. Let S, T, Q occupy respectively z, w, v cosets of H. These are the occupied classes modulo g, and Kneser's theorem gives

$$v \ge z + w - 1. \tag{3.5}$$

Set

$$\alpha = \left\lfloor \frac{l}{g} \right\rfloor, \qquad \beta = \left\lceil \frac{l}{g} \right\rceil. \tag{3.6}$$

Since $l < m = gh$, we have $0 \le \alpha \le \beta \le h$.

Count the elements of I by their classes modulo g. The zero class contains at most $C_B(g)$ elements of I in total. This count includes both parts of (3.2). In each nonzero class, $[l+1, m]_{\mathbb{Z}}$ contains at most $h - \alpha$ elements, since

$$\left\lceil \frac{m - l}{g} \right\rceil = h - \left\lfloor \frac{l}{g} \right\rfloor.$$

There are at most $w - 1$ nonzero classes occupied by the low generators. The high interval $[d+1, k]_{\mathbb{Z}}$ has length l and contains at most β elements in any class; at most $z - 1$ nonzero classes are relevant for its elements. Hence

$$|I| \le C_B(g) + (w-1)(h-\alpha) + (z-1)\beta. \tag{3.7}$$

On the other hand, $0 \in Q$ and Q is H-periodic, so the entire zero coset H is contained in Q. In A this coset contributes $C_A(g) \ge C_B(g)$ elements by (D). Each other occupied coset contributes at least $\lfloor d/g \rfloor \ge h$ elements, because $d \ge m = gh$. All of these elements are representable by (3.4). Consequently

$$|A \cap \langle I \rangle| \ge C_B(g) + (v-1)h \ge C_B(g) + (z+w-2)h. \tag{3.8}$$

The last lower bound minus the upper bound in (3.7) equals

$$(z-1)(h-\beta) + (w-1)\alpha \ge 0. \tag{3.9}$$

This proves the lemma. □

Remark 3.2. The proof does not require the map $I \longrightarrow \mathbb{Z}/m\mathbb{Z}$ to be injective. It counts original generators through their interval capacities, while Kneser's theorem counts occupied cosets. It also allows a nontrivial stabilizer. In fact, after the stabilizer modulus g has been determined, the proof uses (D) only for that one modulus.

4 Subsets near the upper endpoint

We next handle subsets having no generator at most d, provided that $k \geq 3l - 1$. This lemma needs only the existence of a direct multiple for each generator.

Lemma 4.1 (Upper-endpoint subsets)**.** *Let $l \geq 1$, $k \geq 3l - 1$, and $d = k - l$. Let $A = [N, U]_{\mathbb{Z}}$ have length d, with $N > k$. Suppose $\varnothing \neq I \subseteq [d + 1, k]_{\mathbb{Z}}$ and every $b \in I$ has a multiple in A. Then $|A \cap \langle I \rangle| \geq |I|$.*

Proof. Let D_I be the union, over $b \in I$, of the multiples of b in A. If $|D_I| \geq |I|$, the conclusion is immediate. Otherwise, since every generator has at least one such multiple, two distinct generators $a < b$ have a common multiple $x \in A$. Put

$$g = \gcd(a, b), \quad a = ug, \quad b = vg, \quad J = \left\lfloor \frac{k}{g} \right\rfloor, \quad L = \frac{ab}{g}, \quad F = L - a - b. \tag{4.1}$$

Since x is a positive common multiple, $U \geq x \geq L$.

The interval $[d + 1, k]_{\mathbb{Z}}$ has length l. Thus $g \leq b - a \leq l - 1$, and

$$d = k - l \geq 2l - 1 \geq l + g.$$

It follows that

$$\left\lfloor \frac{d}{g} \right\rfloor \geq \left\lceil \frac{l}{g} \right\rceil. \tag{4.2}$$

For example, to see the floor-ceiling step, use $d/g \geq l/g + 1$ and $\lfloor l/g \rfloor + 1 \geq \lceil l/g \rceil$. The collision case is impossible for $l = 1$, so all the expressions above are consistent with $g \geq 1$.

Also $d \geq k/2$, so $u, v > k/(2g) \geq J/2$. The distinct integers $u < v$ therefore satisfy

$$u + v \geq J + 2. \tag{4.3}$$

Indeed, if $J = 2j$, their sum is at least $2j + 3$; if $J = 2j + 1$, it is at least $2j + 3$.

Fix a residue class modulo g occupied by I. Let c be the smallest generator in this class, and let r_c be its number of generators. We have both

$$r_c \leq \left\lceil \frac{l}{g} \right\rceil, \qquad r_c \leq \left\lfloor \frac{k - c}{g} \right\rfloor + 1. \tag{4.4}$$

We will produce r_c points $y \in A$ satisfying $y \equiv c \pmod{g}$ and $y > F + c$. Since $y > k \geq c$, the difference $y - c$ is nonnegative. Lemma 2.3 then gives

$$y \in c + \langle a, b \rangle \subseteq \langle I \rangle.$$

Define

$$\begin{aligned} C_c &= \left\lfloor \frac{U - c}{g} \right\rfloor - \left\lfloor \frac{N - 1 - c}{g} \right\rfloor, \\ Q_c &= \left\lfloor \frac{U - c}{g} \right\rfloor - \frac{F}{g}. \end{aligned} \tag{4.5}$$

Because F/g is an integer, the exact number of selected points is

$$\max\{0, \min(C_c, Q_c)\}. \tag{4.6}$$

For completeness, put $j_{\max} = \lfloor (U - c)/g \rfloor$ and $j_{\min} = \lfloor (N - 1 - c)/g \rfloor + 1$. The points in A in this residue class are $c + jg$ for $j_{\min} \leq j \leq j_{\max}$. The threshold $y > F + c$ imposes $j \geq F/g + 1$. Thus their number is $\max\{0, j_{\max} - \max(j_{\min}, F/g + 1) + 1\}$, which is exactly (4.6).

Every residue class has at least $\lfloor d/g \rfloor$ points in an interval of length d. Equations (4.2) and (4.4) give $C_c \geq r_c$. Using $U \geq L$ and (4.3), we also obtain

$$\begin{aligned} Q_c &\geq u + v - \left\lceil \frac{c}{g} \right\rceil \\ &\geq J + 2 - \left\lceil \frac{c}{g} \right\rceil \\ &\geq \left\lfloor \frac{k-c}{g} \right\rfloor + 1 \geq r_c. \end{aligned} \tag{4.7}$$

The penultimate inequality follows from

$$\left\lceil \frac{c}{g} \right\rceil + \left\lfloor \frac{k-c}{g} \right\rfloor \leq J + 1.$$

One can verify this by writing c/g as its integer and fractional parts and using $\lfloor k/g \rfloor = J$. Hence (4.6) is at least r_c. Summing over the occupied residue classes proves the lemma, since those classes are disjoint. □

Corollary 4.2. *Under the hypotheses of Theorem 1.2, if $k \geq 3l - 1$, then* (H) *holds for every nonempty $I \subseteq B$.*

Proof. If I has an element at most d, apply Lemma 3.1. Otherwise $I \subseteq [d+1, k]_{\mathbb{Z}}$. If $l = 0$, this latter case is empty; if $l \geq 1$, (D) supplies the direct multiples required by Lemma 4.1. □

Remark 4.3. The modulus g in Lemma 4.1 is the greatest common divisor of one collision pair. No divisibility by g is required of the other generators. Each occupied class is treated through its own translate $c + \langle a, b \rangle$, and (4.6) keeps the left endpoint N in the count.

5 At most three generators

This section proves a small-cardinality statement for a more general pair of positive integer intervals. It will supply the initial cases of the iterated-sumset argument.

Lemma 5.1. *Let A, B be nonempty finite intervals of positive integers, with* $\min A > \max B$ *and $|A| \geq |B|$. Assume $|A \cap s\mathbb{Z}| \geq |B \cap s\mathbb{Z}|$ for every positive integer s. Then $|A \cap \langle I \rangle| \geq |I|$ for every $I \subseteq B$ with $1 \leq |I| \leq 3$.*

Proof. **One or two generators.** For $I = \{b\}$, the multiple-count inequality with $s = b$ gives a representable point in A. For $I = \{a, b\}$, consider the union of their direct multiples in A. If it has at least two elements, there is nothing to prove. Otherwise both generators have the same unique direct multiple $x \in A$. Let $g = \gcd(a, b)$ and $L = ab/g$. The interval B has at least two multiples of g, so A does as well. Since the multiples of g in an interval are consecutive along the g-lattice, at least one of $x - g$ and $x + g$ lies in A. That point is positive and at least

$$x - g \geq L - g > L - a - b.$$

It is representable by Lemma 2.3, and is distinct from x.

Reduction for three generators. Let $|I| = 3$, and first divide by the overall greatest common divisor $h = \gcd(I)$. The sets

$$A_h = (A \cap h\mathbb{Z})/h, \qquad B_h = (B \cap h\mathbb{Z})/h$$

are separated intervals of positive integers. They are nonempty, $|A_h| \geq |B_h|$, and their multiple counts for s are the original counts for hs. Also $I/h \subseteq B_h$ has three elements. Any three

representable points of A_h scale back to three such points of A. We may therefore assume $\gcd(I) = 1$.

If the union of all direct multiples has at least three points, we are done. Otherwise there is a collision pair $a < b$ in I with a common multiple $x \in A$; write c for the third generator, and put

$$g = \gcd(a, b), \qquad L = ab/g, \qquad F = L - a - b.$$

If A contains at least three multiples of g, choose three consecutive such multiples containing x. Their smallest is at least $x - 2g \geq L - 2g > F$, since $a + b > 2g$. All three are representable by Lemma 2.3.

It remains that A contains exactly two multiples of g. The multiple-count inequality forces B to contain exactly two as well. Consequently, for some integer $v \geq 1$,

$$a = vg, \qquad b = (v + 1)g, \qquad c < (v + 2)g. \tag{5.1}$$

The first two expressions follow because two multiples in an interval with no third must be consecutive. The last inequality holds because B does not contain the next multiple $(v + 2)g$. Since $\gcd(a, b, c) = 1$, we have $g > 1$ and $c \not\equiv 0 \pmod{g}$. Indeed, $g = 1$ would make all three elements of I multiples of g. We also have

$$|A| \geq |B| \geq g + 1, \qquad \min A > x - 2g \geq L - 2g. \tag{5.2}$$

The last strict inequality holds because $x - 2g$ in A would give three multiples of g. The two multiples of g in A are themselves representable: they are at least $x - g \geq L - g > F$. We seek one further point in the nonzero residue class of c.

The remaining cases. If $v = 1$, then $a = g$. By (5.2), A contains a point congruent to c modulo g; since that point exceeds c, it belongs to $c + \langle g \rangle$.

If $v \geq 3$, then (5.1) gives

$$F + c < L - (v - 1)g \leq L - 2g < \min A.$$

Every point of A in the residue class of c therefore belongs to $c + \langle a, b \rangle$. At least one exists by (5.2).

Suppose now $v = 2$. Then $a = 2g$, $b = 3g$, $L = 6g$, and $F = g$. If $c \leq 3g$, we have $F + c \leq 4g < \min A$, and the same argument applies. If $c > 3g$, write $c = 3g + s$ with $1 \leq s < g$, and $x = 6jg$ with $j \geq 1$. Both integers

$$\begin{aligned} y &= x - g + s = c + 2g(3j - 2), \\ z &= x + s = c + 3g(2j - 1) \end{aligned} \tag{5.3}$$

belong to $\langle I \rangle$ and have the nonzero residue of c. They satisfy $y < x < z$ and $z - y = g$. An integer interval containing x and avoiding both y and z would be contained in $[y + 1, z - 1]_{\mathbb{Z}}$ and have length at most $g - 1$. This contradicts $|A| \geq g + 1$. Thus at least one of y, z belongs to A and supplies the third representable point. All cases are complete. □

6 The small-index criterion

Lemma 6.1. *Under the hypotheses of Theorem 1.2, let $t = \lfloor n/k \rfloor \geq 2$. If $l \leq 3(t - 1)$, then* (H) *holds for every nonempty $I \subseteq B$.*

Proof. If $|I| \leq 3$, apply Lemma 5.1. Henceforth put $r = |I| \geq 4$ and $m = \max I$. If $m \leq d$, move each generator $b \in I$ forward by a nonnegative multiple of m into the interval $[N, N + m - 1]_{\mathbb{Z}}$.

The resulting points are representable and distinct: different elements of $[1, m]_{\mathbb{Z}}$ have different residues modulo m. Since $m \leq d$, the whole target interval lies in A.

We may therefore suppose $m > d$. Put

$$e = m - d \leq l, \qquad S = I \bmod m, \qquad W = A \bmod m.$$

Then $|S| = r$, $|W| = d$, and $0 \in S$. The equalities of cardinalities use $I \subseteq [1, m]_{\mathbb{Z}}$ and $d < m$, respectively.

Lifting the iterated sumset. Since $n \geq tk$,

$$N = n - k + 1 > (t-1)k \geq (t-1)m. \tag{6.1}$$

If $y \in A$ and $y \bmod m \in tS$, choose at most t generators representing its summands, using the empty sum for a zero residue. Every chosen nonzero generator is less than m, so their sum z satisfies $0 \leq z < tm$. If $z > y$, congruence modulo m and (6.1) would give $z \geq y + m > tm$, a contradiction. Thus $y = z + jm \in \langle I \rangle$ for some $j \geq 0$. It remains to show $|W \cap tS| \geq r$.

Counting with the stabilizer. Write $H = \mathrm{Stab}(tS)$ in the form (2.4), and put $h = m/g$. Let S and tS occupy z and w cosets of H. By (2.3),

$$w \geq tz - (t-1) \geq 2z - 1. \tag{6.2}$$

Define

$$v = h - \left\lfloor \frac{l}{g} \right\rfloor.$$

Because $l < m = gh$, we have $v \geq 1$. The interval $[l+1, m]_{\mathbb{Z}}$ contains at most v integers in each class modulo g, so

$$r \leq zv. \tag{6.3}$$

Every coset contains at least $v - 1$ elements of W. Indeed, $d \geq m - l$ implies

$$\left\lfloor \frac{d}{g} \right\rfloor \geq h - \left\lceil \frac{l}{g} \right\rceil \geq v - 1.$$

For the zero coset, (D) gives the stronger bound

$$|W \cap H| = C_A(g) \geq C_B(g) = \left\lfloor \frac{k}{g} \right\rfloor - \left\lfloor \frac{l}{g} \right\rfloor \geq v, \tag{6.4}$$

since $k \geq m$. The equality $|W \cap H| = C_A(g)$ uses the injectivity of reduction of A modulo m.

If $v \geq 2$, then $0 \in tS$ implies $H \subseteq tS$. Using (6.2) and (6.4), we get

$$|W \cap tS| \geq v + (2z-2)(v-1) = zv + (z-1)(v-2) \geq r. \tag{6.5}$$

Suppose $v = 1$. Each occupied coset of S now contains exactly one element of S, so $z = r$. If $d \geq g$, every coset has at least one element of W; hence $|W \cap tS| \geq w \geq r$. If $d < g$, the interval A occupies d distinct classes modulo g. There are g such classes in total, and tS occupies w of them. Therefore

$$|W \cap tS| \geq d + w - g \geq tr - (t-1) - (g-d). \tag{6.6}$$

Finally,

$$(t-1)(r-1) \geq 3(t-1) \geq l \geq m - d \geq g - d,$$

where we used $r \geq 4$, the hypothesis on l, and $m \leq k$. Thus the last expression in (6.6) is at least r. This proves the lemma. □

7 Layer restrictions forced by polynomiality

The next proposition uses only the multiple-count condition. It is the parameter reduction that makes the preceding two ranges sufficient, up to two explicit boundary families.

Proposition 7.1. *Under the hypotheses of Theorem 1.2, suppose $d \geq 2$ and put $t = \lfloor n/k \rfloor \geq 2$. Then*

$$k \geq (t+1)(l-t+1), \tag{7.1}$$

and

$$tk + l \leq n \leq \min\{(t+1)k - 1,\ (t+1)k - tl + t^2 - 1\}. \tag{7.2}$$

Proof. The definition of t gives

$$(t-1)k < N \leq tk, \qquad U < (t+1)k.$$

Since $k \in B$, condition (D) forces a multiple of k in A. The only possible one is tk, so $tk \in A$. As A has length d,

$$N \geq tk - d + 1. \tag{7.3}$$

Now take

$$b = \left\lfloor \frac{N-1}{t} \right\rfloor.$$

Since $N \leq tk$, we have $b \leq k-1$. We claim $b \geq l+1$. Otherwise $b \leq l$ implies $N \leq t(l+1)$. Both $t(l+1)$ and tk would then lie in A: they are between N and $tk \leq U$. Their difference is $t(d-1) > d-1$, which is impossible in an interval of length d. Thus $b \in B$.

By construction $tb < N$. Condition (D) supplies a multiple of b in A, and its multiplier must be at least $t+1$. Hence

$$(t+1)b \leq U = N + d - 1.$$

Write $N = tb + v$ with $1 \leq v \leq t$. The last inequality yields $b \leq d + v - 1 \leq d + t - 1$, and therefore

$$N \leq t(d+t). \tag{7.4}$$

Combining (7.3) and (7.4), and using $d = k - l$, gives

$$tk - (k-l) + 1 \leq t(k-l+t), \qquad k \geq (t+1)l - t^2 + 1 = (t+1)(l-t+1).$$

This proves (7.1). Adding $k-1$ to (7.3) and (7.4) gives the corresponding lower and upper bounds for n. The further bound $n \leq (t+1)k - 1$ is part of the definition of t. Together these are (7.2). □

8 The two boundary families

The relevant boundary parameters have $t = 2$ and k equal to $3l-3$ or $3l-2$. We first determine n, and then give explicit injections for all subsets having no generator at most d.

Lemma 8.1. *Assume the hypotheses of Theorem 1.2, with $l \geq 3$ and $t = 2$.*

(i) *If $k = 3l-3$, then $n = 7l-6$.*

(ii) *If $k = 3l-2$, then $n = 7l-3$.*

Proof. For $t = 2$, equation (7.2) gives

$$2k + l \le n \le 3k - 2l + 3.$$

For $k = 3l - 3$, the two bounds are both $7l - 6$. For $k = 3l - 2$, they give $n \in \{7l - 4, 7l - 3\}$. The first choice violates (D). Indeed, $b = 2l - 1$ belongs to B, whereas for $n = 7l - 4$ one has

$$A = [4l - 1, 6l - 4]_{\mathbb{Z}}, \qquad 2b = 4l - 2 < \min A, \qquad 3b = 6l - 3 > \max A.$$

Since two consecutive multiples of b straddle the whole interval, A contains no multiple of b. Thus only $n = 7l - 3$ is possible. □

Lemma 8.2 (Boundary injections)**.** *Let $l \ge 3$. For either of the two parameter families*

$$(k, n) = (3l - 3, 7l - 6) \quad \text{or} \quad (k, n) = (3l - 2, 7l - 3),$$

every nonempty $I \subseteq [d + 1, k]_{\mathbb{Z}}$ satisfies $|A \cap \langle I \rangle| \ge |I|$. This assertion does not require (D)*.*

Proof. **The family $k = 3l - 3$.** Set $g = l - 1 \ge 2$. The parameters become

$$k = 3g, \qquad d = 2g - 1, \qquad A = [4g + 2, 6g]_{\mathbb{Z}}, \qquad I \subseteq [2g, 3g]_{\mathbb{Z}}.$$

If $3g \in I$, the map

$$c \longmapsto c + 3g \qquad (c \in I) \tag{8.1}$$

is injective and has representable values in $[5g, 6g]_{\mathbb{Z}} \subseteq A$. The inclusion uses $g \ge 2$. If $3g \notin I$, use

$$c \longmapsto \begin{cases} 6g, & c = 2g, \\ 2c, & c > 2g. \end{cases} \tag{8.2}$$

The exceptional value is $3(2g)$. Every other value is an even integer between $4g + 2$ and $6g - 2$. They are mutually distinct and avoid $6g$. Thus (8.2) is also an injection into $A \cap \langle I \rangle$.

The family $k = 3l - 2$. Here

$$d = 2l - 2, \qquad A = [4l, 6l - 3]_{\mathbb{Z}}, \qquad I \subseteq [2l - 1, 3l - 2]_{\mathbb{Z}}.$$

The map

$$c \longmapsto \begin{cases} 3c = 6l - 3, & c = 2l - 1, \\ 2c, & c \ge 2l \end{cases} \tag{8.3}$$

has representable values in A. Its exceptional value is odd, whereas the other values are distinct even integers in $[4l, 6l - 4]_{\mathbb{Z}}$. This proves injectivity and completes both families. □

9 Completion of the proof

Proof of Theorem 1.2. If $d = 1$, then $B = \{k\}$ and $A = \{n - k + 1\}$. Condition (D) with $s = k$ gives $k \mid n - k + 1$, so $A \subseteq \langle k \rangle$ and (H) follows. The corresponding quotient is the finite geometric sum

$$\frac{1 - q^{n-k+1}}{1 - q^k} = \sum_{j=0}^{(n-k+1)/k-1} q^{jk}.$$

Suppose $d \ge 2$, and put $t = \lfloor n/k \rfloor \ge 2$. If $l \le 3(t - 1)$, Lemma 6.1 proves (H) for every nonempty subset of B. If $k \ge 3l - 1$, Corollary 4.2 gives the same conclusion. It remains to suppose that both inequalities fail:

$$l \ge 3t - 2, \qquad k < 3l - 1. \tag{9.1}$$

Proposition 7.1 gives

$$\begin{aligned} k-(3l-1) &\geq (t-2)l - t^2 + 2 \\ &\geq (t-2)(3t-2) - t^2 + 2 \\ &= 2(t-1)(t-3). \end{aligned} \tag{9.2}$$

The second inequality uses $t \geq 2$. If $t \geq 3$, the last expression is nonnegative, contradicting $k < 3l-1$. Hence $t = 2$. Equation (7.1) now says $k \geq 3l-3$, so the only integer choices are

$$k = 3l-3 \quad \text{or} \quad k = 3l-2. \tag{9.3}$$

Moreover $l \geq 4$ by (9.1). Lemma 8.1 supplies the corresponding values of n.

Fix any nonempty $I \subseteq B$ in either remaining family. If it contains a generator at most d, apply Lemma 3.1. Otherwise apply Lemma 8.2. Thus every such I satisfies (H). The cases above cover all normalized parameters, proving Theorem 1.2. □

Parameter range	**Argument giving (H) for all subsets**
$d = 1$	The unique denominator generator divides the numerator exponent
$d \geq 2,\ l \leq 3(t-1)$	Lemma 6.1
$d \geq 2,\ k \geq 3l-1$	Mixed sumsets and the upper-endpoint lemma
Neither preceding inequality, $d \geq 2$	$t = 2$ and the two boundary families; mixed sumsets and explicit injections

Proof of Theorem 1.1. Reduce to $0 \leq l < k \leq n/2$ as in Section 1. Polynomiality gives (D) by (1.6). Theorem 1.2 gives (H) for every nonempty $I \subseteq B$. Applying Theorem 2.1 to (1.5) yields $P_{n,k,l}(q) \in \mathbb{Z}_{\geq 0}[q]$. The equal-index cases have quotient 1, and the other ordering cannot give a polynomial. This proves the full statement. □

Corollary 9.1 (A Hilbert-series realization)**.** *Every polynomial quotient in the normalized range of Theorem 1.1 is the Hilbert series of a finite-dimensional graded complete intersection*

$$\mathbb{C}[x_{l+1}, \ldots, x_k]/(f_N, \ldots, f_U), \qquad \deg x_b = b, \quad \deg f_a = a.$$

Proof. Theorem 1.2 verifies the hypothesis of the regular-sequence assertion in Theorem 2.1. □

Appendix. AI use and supplementary verification

Author and AI use. Justin Du supplied the candidate proof and verification materials and directed the preparation of this preprint. The manuscript was prepared in an AI-assisted workflow. OpenAI Codex was used to review the supplied argument, expand its mathematical exposition, check references, rerun the verification programs, and prepare the LaTeX source and PDF. The author is responsible for the preprint and is the contact for questions and corrections.

Review status. The argument has undergone internal AI-assisted mathematical review and exact-arithmetic checks. External human peer review and proof-assistant formalization are not supplied with this version. The word *candidate* records that review status. All parameter branches are addressed in the text, relative to the two external inputs explicitly identified in Section 2. This version makes no claim of bibliographic priority.

Role of the computations. The proof of Theorems 1.2 and 1.1 is the symbolic argument in Sections 1–9. No computational cutoff or random sample is used as a premise. The supplied programs test the constructions, capacity inequalities, and exact coefficient identities on finite sets of inputs. Algorithmic cross-checks are distinct from external mathematical review.

Main audit. The program `verify_mixed_completion.py` scans every triple (k, l, n) with

$$2 \le k \le 16, \qquad 1 \le l < k, \qquad 2k \le n \le 7k.$$

It tests (D) and, for each polynomial instance, checks every nonempty subset of B. The randomized stage accepts 6,000 polynomial parameter draws with $15 \le k \le 600$ and tests selected subsets, including subsets with few low generators and subsets confined to the upper endpoint. These draws need not be distinct. The two boundary injections are also checked for $3 \le l \le 300$ without imposing polynomiality. The coefficient checks multiply each selected full quotient back by the denominator and compare with the complete numerator polynomial, without truncating the product identity.

Main audit item	**Exact count**
Parameter triples in the exhaustive scan	6,920
Polynomial triples in that scan	971
Random accepted polynomial draws	6,000
Subset checks, including selected random subsets	3,745,276
Mixed-sumset checks	3,726,163
Mixed checks with a nontrivial stabilizer	2,399,416
Mixed checks with collisions under reduction modulo m	517,086
Comparisons of two semigroup algorithms	1,494
Boundary-injection checks without polynomiality	4,768
Complete polynomial identities	58
Quotient coefficients in those identities	82,340

The main semigroup oracle uses a bitset and successive doubling of the available number of each generator. A second implementation computes membership by the recurrence $u_0 = 1$ and $u_y = \max_{b \in I,\ b \le y} u_{y-b}$, taking the maximum of an empty set to be 0. The 1,494 comparisons between these implementations all agree. For a constructed certificate, the program checks that its points actually belong to the independently computed semigroup. For subsets of cardinality at most three in the small-index routine, the code tests the conclusion by the oracle; the paper proof of those cases is Lemma 5.1.

Local mixed-sumset audit. The program `audit_local_mixed.py` does not impose full polynomiality. It generates 12,000 inputs with $k \le 2{,}500$, finds the actual stabilizer modulus g of the mixed sumset, and tests only the local hypothesis $C_A(g) \ge C_B(g)$ used in Remark 3.2. Exactly 9,840 generated inputs satisfy this hypothesis and all pass. The other 2,160 are recorded as not meeting the hypothesis. Of the passing inputs, 176 have a nontrivial stabilizer and 9,664 have a trivial stabilizer. An additional semigroup-oracle check is performed on 246 of them.

Reproducibility. The original supplementary archive and its scripts are preserved in the source package. The following commands require Python 3.10 or newer and its standard library:

```
python verify_mixed_completion.py -k-max 16
  -random-cases 6000 -seed 202611103
  -out audit_main_rerun.json
python audit_local_mixed.py -cases 12000
  -seed 202611102 -out audit_local_rerun.json
```

Each displayed command is entered as one line. Both commands were rerun while preparing this version. Their output agrees with the supplied audit files in every field except elapsed time. All nine hashes in the supplied `SHA256.json` were verified; the standalone proof draft and the draft inside the original archive are byte-for-byte identical. The supplement also records a counterexample to an unrestricted double-sumset shortcut; that rejected assertion is not used in this paper.

References

[1] Mona Gatzweiler and Christian Krattenthaler. *A positivity conjecture for a quotient of q-binomial coefficients*. The Ramanujan Journal **69** (2026), article 13, 13 pp. DOI: 10.1007/s11139-025-01285-2. Preprint: `https://arxiv.org/abs/2502.06032`.

[2] Mona Gatzweiler, Fabián Levicán-Santibáñez, and Atsuro Yoshida. *Cyclotomic generating functions, empty weighted complete intersections and positivity*. Preprint, arXiv:2603.22226v1 (2026). `https://arxiv.org/abs/2603.22226v1`.

[3] Matt DeVos. *A short proof of Kneser's addition theorem for abelian groups*. Preprint, arXiv:1303.3539v1 (2013). `https://arxiv.org/abs/1303.3539v1`.